

Explanatory Notes

Devices X4, 5, 13 and X12 multiplex the 14 bit address and the the 7 bit refresh address busses onto the 7 bit RAM address bus.

The multiplexer control signals (X11 pin 5 and X10 pin 9) select the CPU address bus and the refresh address bus respectively.

B $\overline{\text{OE}}$, on X9 pin 13 conditions the first part of each OE cycle for refresh, if required. The remaining part of the cycle is reserved for normal read/write operations.

RAS the row address select occurs twice per OE cycle since it is required for both refresh and data read/write.

CAS - the construction of this waveform includes the chip select signal (X10 pins 16 and 9). It is not required during refresh and therefore occurs only during the latter half of a OE cycle.

Additional address decoding

Pads are provided on X18 pins 2-6 to be wired as required to any appropriate edge connector. These inputs may be used to provide an extension to the normal address space (64K) or to disable part of a 16K block that may already be used for ROM, RAM or I/O. *By bringing any of these pins low the memory block will be deselected.*
Write protection

One 16K byte block may be 'protected' by bringing X18 pin 9 low. *Pin 2 on diagram.*
Pads are provided to allow strapping to an edge connector or a switch.

Address Selection

switch from '1' is further from the edge conn.

X10 SWITCH POSITION	ADDRESSES X23-X30	X10 SWITCH POSITION	ADDRESSES X31-X38
1	0000-3FFF	2	0000-3FFF
3	4000-7FFF	4	4000-7FFF
5	8000-BFFF	6	8000-BFFF
7	C000-FFFF	8	C000-FFFF

Components DYNARAM

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X12, X13, X4, X5	74LS153
X23-X38	MB4116 (NE or H)
X20, X21	DM81LS97 <i>or DM81LS95</i>
X9	74LS139 4013
X1, X8, X16	74LS04
X2, X22, X11	74LS74
X19	74LS32 ✓
X14, X6, X7, X15	74LS163
X17	74LS30
X18	74LS00 ✓
D1	5.6V Zener <i>diode</i>
TR1	BCY70
R1, R2, R3, R4, R5, R6, R8, 9, 10	2.7K (TR4)
R7	30.
C1	ELECTROLYTIC 100uf 10V
C2	100uf 20V
X10 <i>switch</i>	8 Pole single throw DIP
All other capacitors (31 total)	0.1uf ceramic disc